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10/3/2025
CS-361
Homework 1 - Theory
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Question 1

a. A Scenario where the final value is n = 10 is where both P and Q would read the old same n value before either P and Q write.

Steps	Threads	Line	Effect	N value after
1	P	p2	P reads n -> temp = 0	0
2	Q	q2	Q reads n -> temp = 0	0
3	P	p3	n = 0 + 1	1
4	Q	q3	n = 0 + 1 (overwriting P)	1
Previous 2 Lines would Repeat another 9 Times	Q	q3	n = 9+1 (overwriting P)	10

b. There wouldn't be a final scenario where the final value is n = 2. To start, in the instructions it says at the very top of the file it says "all files are assumed to be atomic statements. Executions will never context switch in the middle of completing a numbered line." So with this in mind, we only overwrite after each other. So if there NO overwrites it means the final value for would equal = 20. And if there were overwrites the minimum value would be n = 10.

Steps	Threads	Line	Effect	N value after
1	P	p2	P reads n -> temp = 0	0

2	Q	q2	Q reads n -> temp = 0	0
3	P	p3	n = 0 + 1	1
4	Q	q3	n = 0 + 1 (overwriting P)	1
Previous 2 Lines would Repeat another 9 Times	Q	q3	n = 9+1 (overwriting P)	10

Question 2

a. Write a Scenario Table showing this algorithm does not work
   The Scenario would be that P would evaluate f(i) == 0 would be True, but then Q's next evaluation for f(j) == 0 would be False.

Step	Thread	Line	Effect	found After
1	P	p4	P writes found = True	true
2	Q	q4	Q writes found = false	false

b. Write a Scenario Table showing this algorithm does not work
   The Scenario would be that P is f(i) == 0 to be True, then Q's next check would be False.

Step	Thread	Line	Effect	found After
1	P	p3	P writes found = True	true
2	Q	q3	Q writes found = false	false

c. Write a Scenario Table showing this algorithm does not work

Even though the final result would be true it’s still wrong because P holds the value for I and Q wouldn’t be able to see it. So we wouldn’t know for certain what the correct value of i is.

Step	Thread	Line	Effect	found After
1	P	p2	await turn==1 then turn=2	false
2	P	p3	I = i + 1	false
3	P	p4	If f(i) == 0 -> true	false
4	P	p5	Found = true	true
5	Q	q2	await turn==2 then turn=1	true
6	Q	q3	j = j-1	true
7	Q	q4	If f(j) == 0 -> False	true
8	(both)	while-check	Next loop conditions sees if found==true	true